

Harsath Kaliappa Gounder Thangavel

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Software Engineer with 1.5+ years of professional experience and 7+ years of coding experience. Active open-source contributor since 2020, experienced in backend, data engineering, AI integration, and cloud computing.

Education

York University

Bachelor of Science with Honours, Computer Science

Distributed Systems, Computer Networking, Data Mining, Databases, Machine Learning, Operating Systems, DSA, OOP

Toronto, Canada

January 2020 – December 2024

Professional Experience

Software Developer

Privex Inc (privex.io)

October 2023 – January 2024

London, United Kingdom (remote)

- Developed an AI-driven customer support assistant using Retrieval-Augmented Generation (RAG) on a self-hosted LLM, delivering personalized responses to 800+ customers about support tickets, products, and services.
- Reduced support staff workload by 80% while providing instantaneous, context-aware assistance that improved overall customer experience.
- Implemented REST APIs to interface with the LLM and a Redis-based asynchronous task queue for processing customer queries and staff updates via Discord, prioritizing high throughput and robustness under heavy load.
- Designed data ingestion workflows across multiple channels, including support tickets, website documentation, and Discord announcements.
- Built the infrastructure using MySQL for chunked data storage and ChromaDB for embeddings.
- Tech Stack:** Python (Flask, Chroma, Celery, Langchain), Redis, MySQL, Docker, Linux

Machine Learning Engineer

CogniPet AG

April 2019 – December 2019

Zurich, Switzerland (remote)

- Developed a Pet Breed Detection system using Xception neural network and transfer learning, achieving 94% accuracy across 120 dog breeds.
- Implemented REST APIs for model inference and integrated them into CogniPet's Android and iOS apps using AWS EC2.
- Built data management workflows for user-uploaded pet images in AWS S3 with GDPR compliance (timed deletion + transparency).
- Explored video-based techniques for pet facial recognition through multi-frame feature extraction analysis.
- Tech Stack:** Python (Flask, Keras, Pandas, NumPy), AWS (EC2, S3)

Computer Vision Engineer

Gnanamani Ventures Pvt. Ltd.

May 2019 – August 2019

Namakkal, India (on-site)

- Researched and developed a facial recognition-based attendance system using the FaceNet architecture on NVIDIA Jetson Nano, optimizing for edge computing performance.
- Developed data pipelines to pre-process and fine-tune video data of 30+ staff members at Gnanamani College of Technology, resulting in a successful MVP that automated attendance tracking.
- Tech Stack:** Python (TensorFlow, NumPy), NVIDIA cuDNN, OpenCV, Microsoft Excel

Projects

[WikiScroll.xyz](#) | JavaScript, React, Flask, Docker, PostgreSQL

- TikTok-style app for exploring Wikipedia articles with infinite scrolling, bookmarking, and topic-based feeds.
- Used React for frontend, Flask for backend REST APIs, and PostgreSQL for data storage.

[SOCKS5 Handler](#) | C++17, CMake, Networking

- Cross-platform C++ library abstracting SOCKS5 proxy communication with async event loop for efficient TCP stream handling, gaining 32+ GitHub stars.
- Implemented RFC 1928/1929 compliant features including remote DNS resolution, IPv4/IPv6 support, and client authentication.

[Jix Interpreter](#) | C, CMake, GitHub Actions

- Interpreter for a dynamically typed programming language in C.
- Implemented fault-tolerant parsing and error handling, arrays, function pointers, loops, nested blocks, recursion, built-in functions, conditional statements, and control flow statements.
- Implemented functional programming support, including first-class and higher-order functions.

Technical Skills

Languages: Python, Java, JavaScript, C++, C, Bash

Technologies: Linux, Docker, SQL (MySQL, Postgres), NoSQL (MongoDB, Redis, Neo4j), Kafka, AWS

Frameworks: Node.js, React.js, Flask, LangChain, PyTorch, Keras

Tools: Git, Pandas, CI/CD (Github Actions, Jenkins), Build tools (CMake, Maven, Parcel), Jira